

3.9 Organic Synthesis

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Medium

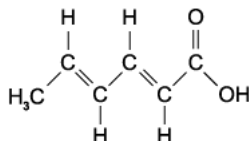
Time allowed: 20

Score: /10

Percentage: /100

Question 1

Sorbic acid is a natural organic compound which is used as a food preservative because it kills fungi. Sorbic acid reacts with bromine in an organic solvent and hydrogen in the presence of a nickel catalyst.



Sorbic acid

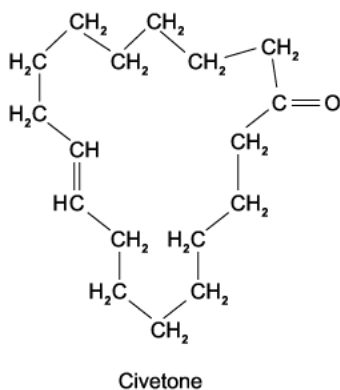
How many moles of bromine (in an organic solvent) and of hydrogen (in the presence of a Pt / Ni catalyst) will be incorporated into one mole of sorbic acid by these reactions?

	Moles of bromine	Moles of hydrogen
A	2	2
B	2	3
C	$2\frac{1}{2}$	2
D	$2\frac{1}{2}$	3

[1 mark]

Question 2

The naturally-occurring molecule civetone was one of the first known ingredients used in perfumery.



Which reagent will **not** react with civetone?

- A. Fehling's reagent
- B. Hydrogen bromide
- C. 2,4-dinitrophenylhydrazine reagent
- D. Sodium tetrahydridoborate (III), NaBH₄

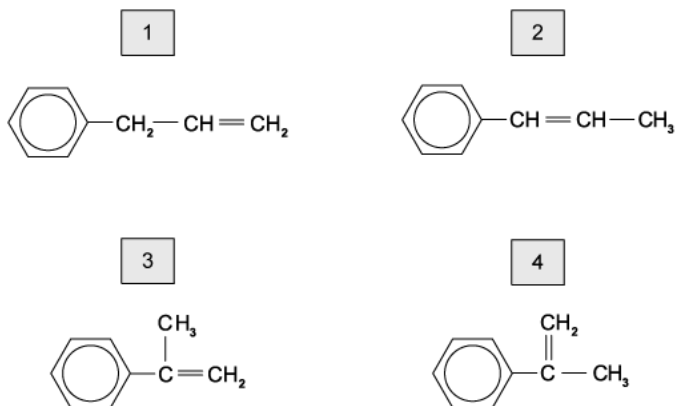
[1 mark]

Question 3

Investigation of an unknown compound, **X**, was carried out in a laboratory and the following properties were discovered:

- **X** has the molecular formula $\text{C}_9\text{H}_{12}\text{O}$;
- **X** is unreactive towards mild oxidising agents.

What is the structure of the compound formed by dehydration of **X**?

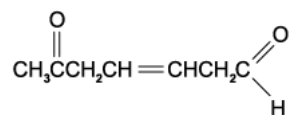


- A. 3 only
- B. 1 and 2
- C. 1, 2 and 3
- D. 3 and 4

[1 mark]

Question 4

A student carried out a series of tests on the compound shown in the diagram below.



Which pairs of reagents would **both** give a positive result for this compound?

- 1 Sodium carbonate and 2,4-dinitrophenylhydrazine reagent
- 2 Fehling's reagent and aqueous bromine
- 3 Tollens' reagent and a solution containing acidified dichromate (VI) ions

A. 1 only

B. 1 and 2

C. 2 and 3

D. 1, 2 and 3

[1 mark]

Question 5

1, 2 and 3 give the molecular formula of organic compounds.

Which would cause decolorisation of aqueous bromine and react with sodium to evolve hydrogen gas?

- 1 C_3H_6O
- 2 C_3H_8O
- 3 $C_3H_4O_2$

A. 1, 2 and 3

B. 1 and 3

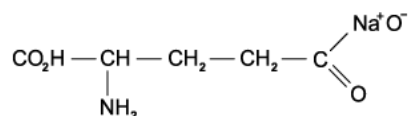
C. 2 and 3

D. 2 only

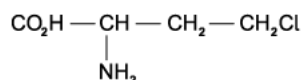
[1 mark]

Question 6

The structure of monosodium glutamate (MSG) is shown below. This molecule is used as a flavour enhancer in foods.



It may be prepared starting from the following compound.



Which set of reagents and reaction conditions could be used to prepare monosodium glutamate?

- 1 Heat under reflux with NaOH (aq)
- 2 Heat with sodium methanoate, $\text{HCO}_2^-\text{Na}^+$
- 3 Heat under reflux with ethanolic KCN followed by hydrolysis with NaOH (aq)

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

Question 7

An alkene is reacted with acidified manganate (VII) ions, MnO_4^- . The organic product has a relative molecular mass greater than that of the alkene by 34.

What are the specific conditions for this reaction?

- A. Cold, dilute MnO_4^-
- B. Cold, concentrated MnO_4^-
- C. Hot, dilute MnO_4^-
- D. Hot, concentrated MnO_4^-

[1 mark]

Question 8

Pentanoic acid can be produced from 1-bromobutane using reagents Y and Z as shown below.



What could be reagents Y and Z?

	Y	Z
A	NaOH (aq)	HCl (aq)
B	NH ₃ in ethanol	H ⁺ /Cr ₂ O ₇ ²⁻ (aq)
C	KCN in ethanol	HCl (aq)
D	KCN in ethanol	NaOH (aq)

[1 mark]

Question 9

In which reaction has a carbocation intermediate?

- A. $\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \rightarrow \text{CH}_3\text{CH}_2\text{OH} + \text{NaBr}$
- B. $\text{CH}_2=\text{CH}_2 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$
- C. $\text{CH}_3\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl} + \text{HCl}$
- D. $\text{CH}_3\text{CHO} + \text{HCN} \xrightarrow{\text{CN}^-} \text{CH}_3\text{CH}(\text{OH})\text{CN}$

[1 mark]

Question 10

Hept-4-enal naturally occurs in cow's milk.



hept-4-enal

What organic product is formed when hept-4-enal is reduced with **either** hydrogen with a nickel catalyst **or** sodium borohydride?

- A. $\text{CH}_3(\text{CH}_2)_5\text{CH}_3$ when hept-4-enal is reduced by H_2 / Ni
- B. $\text{CH}_3(\text{CH}_2)_5\text{CHO}$ when hept-4-enal is reduced by NaBH_4
- C. $\text{CH}_3\text{CH}_2(\text{CH})_2(\text{CH}_2)_3\text{OH}$ when hept-4-enal is reduced by H_2 / Ni
- D. $\text{CH}_3\text{CH}_2(\text{CH})_2(\text{CH}_2)_3\text{OH}$ when hept-4-enal is reduced by NaBH_4

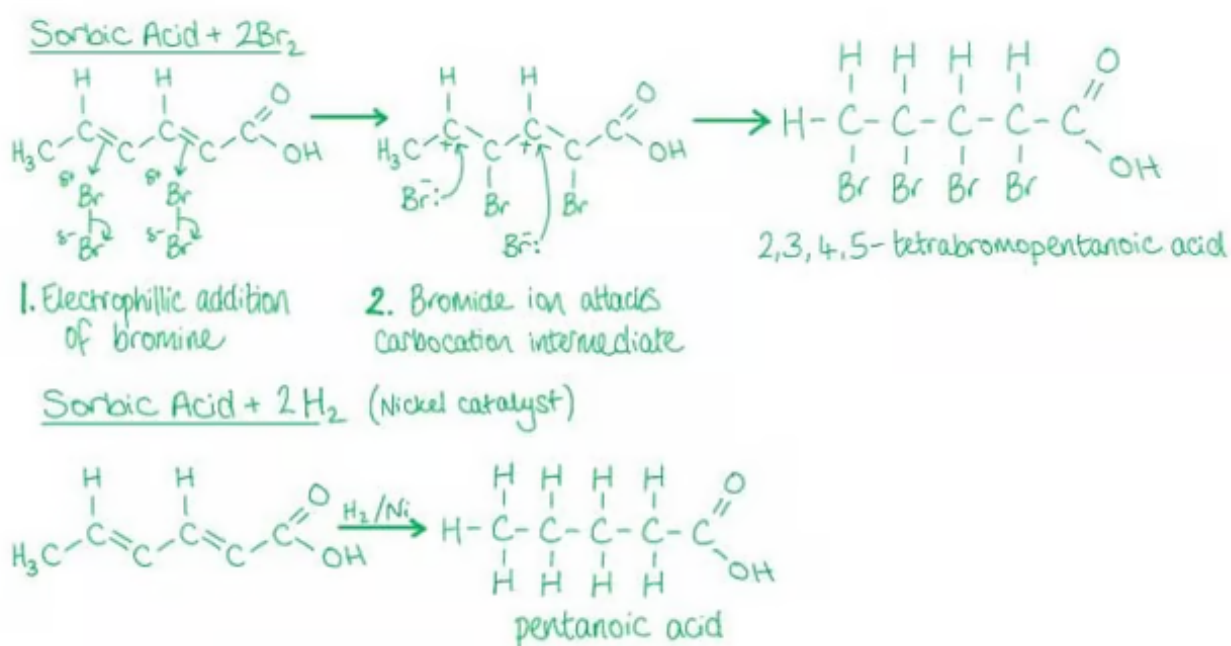
[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- The alkenes are rich in electrons and will be attracted to **electrophiles**.
- The unsaturated C=C will therefore undergo an **electrophilic addition** reaction when shaken with bromine water (Br₂).
 - The orange/yellow colour of bromine will decolourise when reacted with Sorbic acid.
- Since there are two moles of C=C bonds, two moles of Br₂ will react with Sorbic acid.
- Similarly, the two moles of C=C bonds will react with two moles of hydrogen in an **electrophilic addition** reaction.
 - The addition of H₂ to a double bond in the presence of a Pt / Ni catalyst is also called a **hydrogenation** reaction.



The carboxylic acid can undergo acid/base reactions (with alkalis, reactive metals and

carbonates), condensation reacts to form esters or they can react with **reducing agents** to form primary alcohols.

2

The correct answer is **A** because:

- Fehling's reagent tests for the presence of **aldehydes** by oxidising it to a **carboxylic acid**.
 - The Cu^{2+} ions are reduced to Cu^+ ions changing the clear **blue** solution to an **opaque orange/red** due to the formation of copper (I) oxide.
- Civetone does not contain any aldehyde groups, therefore will **not** react with Fehling's reagent.

B is incorrect as the alkenes can undergo **nucleophilic addition** with HBr .

C is incorrect as the ketone group in Civetone can undergo a **condensation reaction** with **2,4-dinitrophenylhydrazine** when heated and form a **deep-orange** precipitate and water.

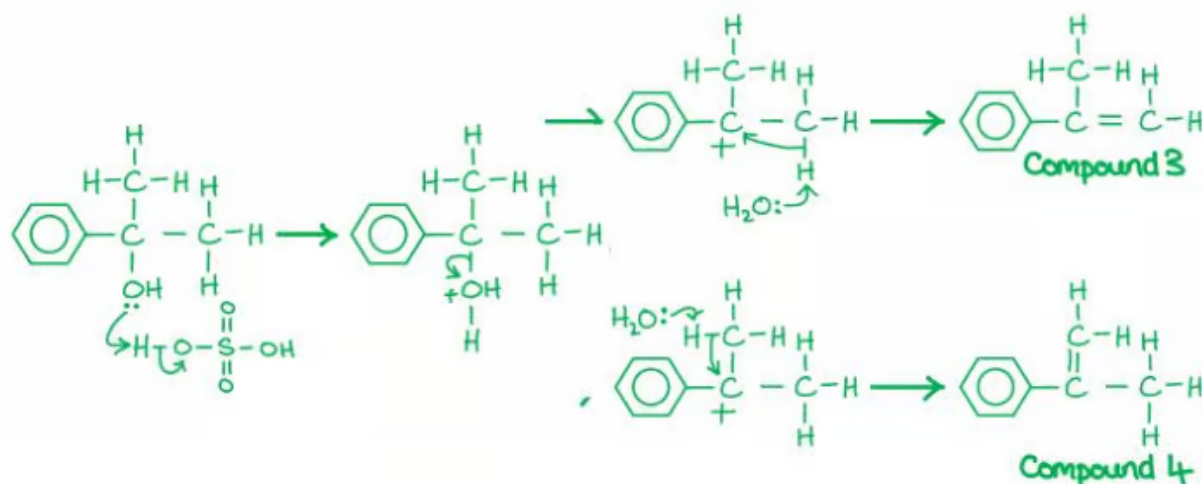
D is incorrect as the ketone group in Civetone will get reduced by NaBH_4 to a secondary alcohol.

3

The correct answer is **D** because:

- Since compound **X** is unreactive towards mild oxidising agents, it is a **tertiary alcohol**.
 - The carbon bonded to the $-\text{OH}$ group is bonded to three other **alkyl** groups.
- A dehydration reaction is an **elimination** reaction in which H_2O is lost.
 - Alcohols undergo dehydration reactions to form an alkene.
 - The alcohol vapour is passed over a hot catalyst of Al_2O_3 powder.
 - Concentrated acid also catalyses the reaction.
- The hydrogens on either alkyl group of the tertiary carbocation can be deprotonated by a water molecule.
- Therefore, a **mixture** of compounds **3** and **4** can be formed.

- Compounds **3** and **4** are the same compounds but drawn differently!



A is incorrect as deprotonation can take place on either alkyl group of the carbocation to produce a **mixture** of compounds **3** and **4**. In this case, both compounds are the same but drawn differently!

B & C are incorrect as the dehydration of a **secondary alcohol** would have produced a **mixture** of compounds **1** and **2**. Secondary alcohols can be oxidised to ketones.

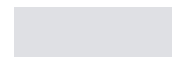
4

The correct answer is **C** because:

- Fehling's reagent tests for the presence of **aldehydes** by oxidising it to a **carboxylic acid**.
 - The Cu^{2+} ions are reduced to Cu^+ ions changing the clear **blue** solution to an **opaque orange/red** due to the formation of copper (I) oxide.
- Orange/yellow** bromine water ($\text{Br}_2(\text{aq})$) will undergo **addition** reactions when shaken with alkenes and turn **colourless**.
- Tollens' reagent tests for the presence of **aldehydes** by oxidising it to a **carboxylic acid**.
 - The Ag^+ ions themselves are reduced to Ag atoms forming a silver 'mirror' on the inside of the tube.
- Acidified dichromate (VI) ions **oxidise** primary alcohols, secondary alcohols and aldehydes.
 - The **orange** dichromate ($\text{Cr}_2\text{O}_7^{2-}$) ions are reduced to **green** Cr^{3+} ions.

- The above compound contains an **aldehyde group** (so it reacts with Fehling's reagent, Tollens' reagent and acidified dichromate ions) and an **alkene group** (so it reacts with aqueous bromine).

A, B & D are incorrect as **aldehydes** and **ketones** react with 2,4-dinitrophenylhydrazine (2,4-DNPH) forming a **deep-orange** precipitate. Sodium carbonate reacts with carboxylic acids to form a salt, water and carbon dioxide. The above compound does not contain any carboxylic acid groups, so the first statement is incorrect.



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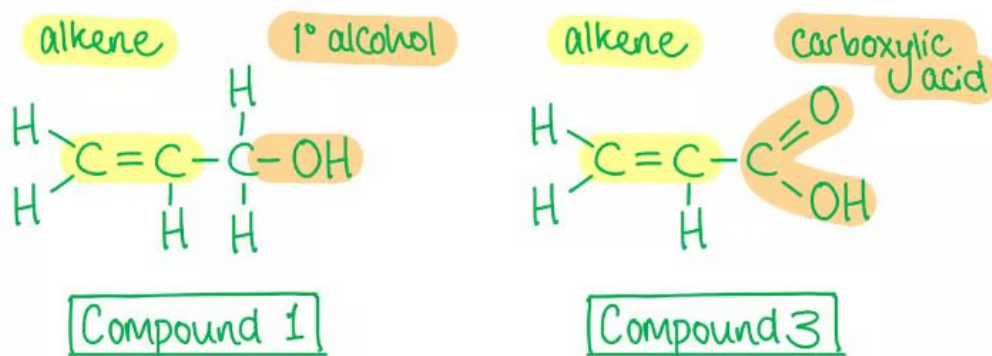
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 - The **orange** dichromate ($\text{Cr}_2\text{O}_7^{2-}$) ions are reduced to **green** Cr^{3+} ions.
- The above compound contains an **aldehyde group** (so it reacts with Fehling's reagent, Tollens' reagent and acidified dichromate ions) and an **alkene group** (so it reacts with aqueous bromine).

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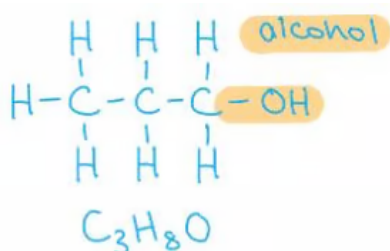
5

The correct answer is **B** because:

- **Alkenes** will undergo an addition reaction when shaken with orange/yellow bromine water ($\text{Br}_2(\text{aq})$) to turn the solution colourless.
- Both alcohols and carboxylic acids will react with reactive metals such as sodium to evolve hydrogen gas.
- Compound **1** contains an **alcohol** and **alkene** group.
- Compound **3** contains a **carboxylic acid** and **alkene** group.



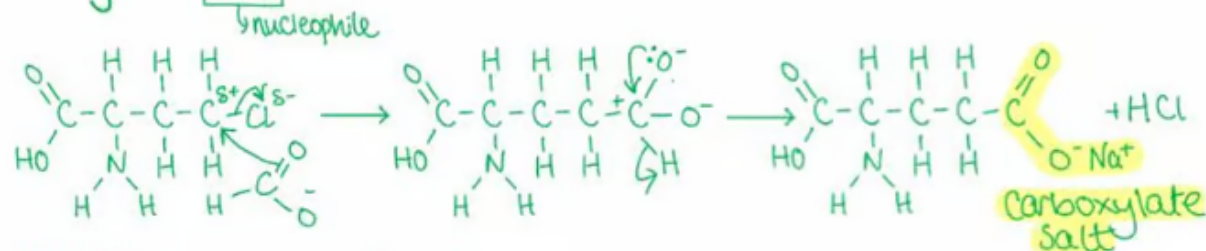
A, C & D are incorrect as compound **2** only contains an **alcohol** group but not **alkene** group.



6

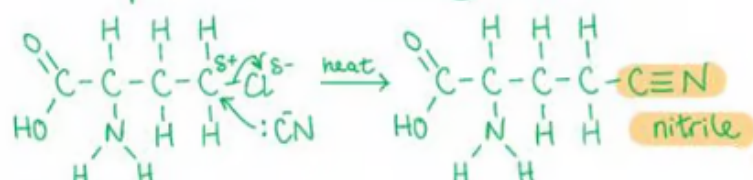
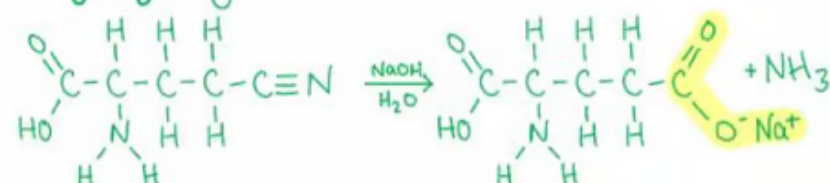
The correct answer is **C** because:

- When heated with $\text{HCO}_2^-\text{Na}^+$ **nucleophilic substitution** of the C/atom takes place by the HCO_2^- which acts as a nucleophile.
 - This produces a carboxylate ion.
- When heated under reflux with ethanolic KCN, **nucleophilic substitution** of the C/ group takes place to form a **nitrile**.
- Hydrolyzing the nitrile will form a **carboxylate salt** (MSG) and **ammonia**.

Heating with $\text{HCO}_2^- \text{Na}^+$ 

KCN/ethanol under reflux then NaOH

1. Nucleophilic Substitution by KCN

2. Hydrolysis by NaOH(aq) 

A, B & D are incorrect as heating under reflux with NaOH(aq) will give **nucleophilic substitution** of the C/atom by the OH^- to form an **alcohol**.

7

The correct answer is **A** because:

- A relative molecular mass of 34 corresponds to two hydrogen and two oxygen atoms.
- Alkenes will undergo mild oxidation by cold, dilute acidified MnO_4^- to form a diol.
- A diol is when an organic compound contains two -OH groups.
- The relative molecular mass of these is $2 \times (16 + 1) = 34$

Alkenes undergo stronger oxidation with **hot, concentrated MnO_4^-** in which the $\text{C}=\text{C}$ fully breaks to form a ketone or aldehyde.

The aldehyde can be further oxidised to a carboxylic acid or CO_2 (if it is formaldehyde, CH_2O).

8

The correct answer is **C** because:

- Halogenoalkanes can undergo **nucleophilic substitution** with ethanolic KCN in which the CN^- ion acts as a nucleophile and replaces the bromine atom in 1-bromobutane to form a **nitrile**.
- The treatment of nitriles with concentrated hydrochloric acid will produce a

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- A relative molecular mass of 34 corresponds to two hydrogen and two oxygen atoms.
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- A diol is when an organic compound contains two $-\text{OH}$ groups.
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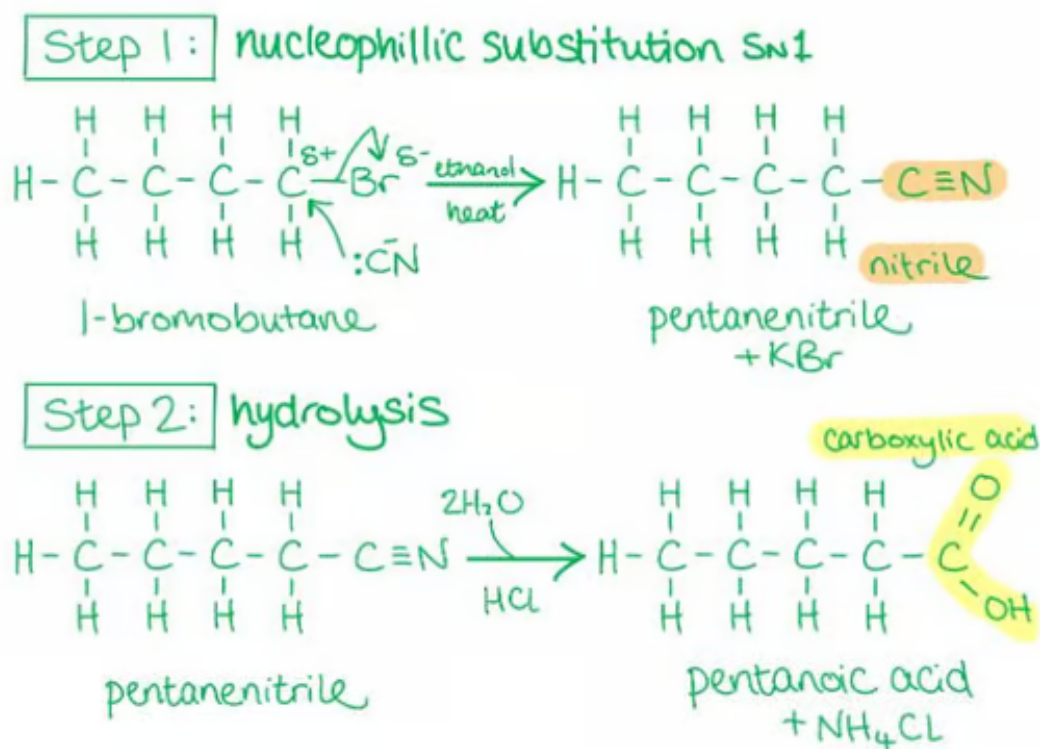
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The correct answer is **C** because:

- Halogenoalkanes can undergo **nucleophilic substitution** with ethanolic KCN in which the CN^- ion acts as a nucleophile and replaces the bromine atom in 1-bromobutane to form a **nitrile**.
- The treatment of nitriles with concentrated hydrochloric acid will produce a carboxylic acid and an ammonium salt.
- In this case, pentanoic acid and ammonium chloride will be formed.



A is incorrect as nucleophilic substitution of Br by OH^- followed by substitution of OH by C^- .

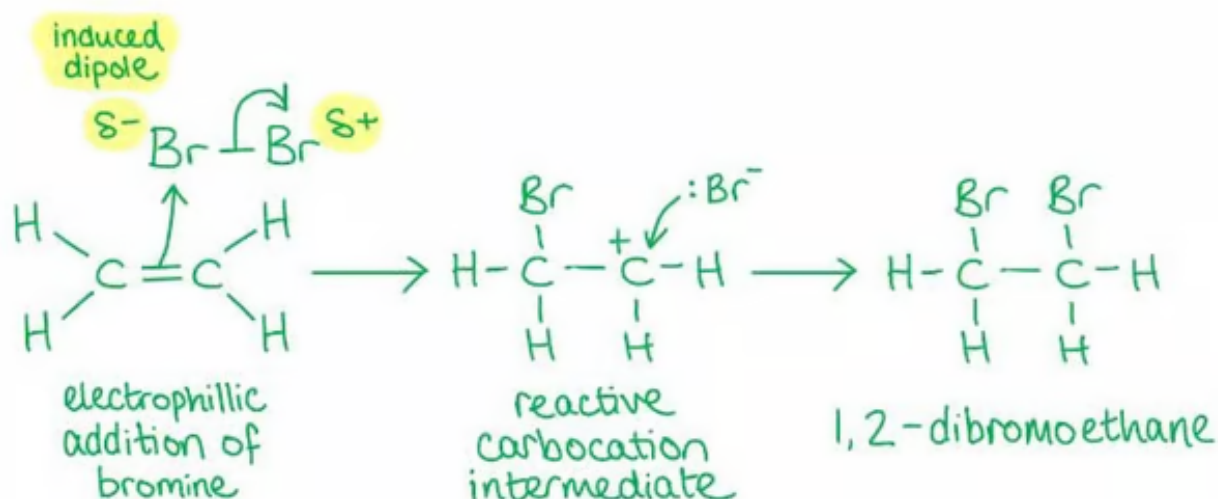
B is incorrect as a nucleophilic substitution reaction takes place to form a primary amine cannot get further oxidised.

D is incorrect as nucleophilic substitution of Br by CN^- followed by substitution of CN by OH^- .

9

The correct answer is **B** because:

- **Alkenes** will undergo an addition reaction when shaken with orange/yellow bromine water (Br_2) to turn the solution colourless.
- The first step involves **electrophilic** attack of the slightly positively charged bromine atom on the $\text{C}=\text{C}$ bond.
- This results in the formation of a primary carbocation which will undergo **nucleophilic** attack by the Br^- to form 1,2-dibromoethane.



A is incorrect as this **nucleophilic substitution** of the bromine by OH^- in the primary halogenoalkane following $\text{S}_{\text{N}}2$ mechanism. This reaction doesn't involve a carbocation intermediate.

C is incorrect as this is a free-radical substitution reaction using sunlight which involves free radicals and not carbocation intermediates.

D is incorrect as this is **nucleophilic substitution addition** of CN^- to the carbonyl carbon to form a hydroxynitrile. This reaction does not involve a carbocation intermediate.

10

The correct answer is **D** because:

- The addition of H_2 to a double bond in the presence of a Pt / Ni catalyst is also called a **hydrogenation** reaction.
- NaBH_4 is a reducing agent, reducing ketones, aldehydes and carboxylic acids to secondary, primary and primary alcohols respectively.
- When reacted with H_2 / Ni, the alkene group in hept-4-enal will undergo hydrogenation.
- When reacted with NaBH_4 the aldehyde group in hept-4-enal will get reduced to a primary alcohol.

